Data Mining Individual Project: A case study on Kickstarter

1.1. Preprocessing & Model Selection

To predict whether a project will succeed or fail, I first cleansed the dataset to remove all project for which *state* is neither "successful" or "failure" as these are not relevant projects for the purposes of this model. Next I removed, unique identified which would be inaccessible/unknown at the initial project launch such as *project ID, pledged, spotlight, staff pick, backers_cout* and the *state* changes. After running a LASSO and permutation importance tests to determine feature predictive power, I removed the fields related to *created,* as they contained no predictive power. Finally, I tested suspected co-linearity between *country* and *currency, blurbl_len* and *blurb_len_clean, name_len* and *name_len_clean.* Only *country, blurb_len* and *name_len* were kept as they offered better predicting power as per LASSO testing.

Next, I converted *goal* to *goal_inUS* using *static_usd_rate* to standardize the goal amount. Without this, the "*country" "currency"* and *"goal"* fields were dependent, decreasing model performance. Lastly, I dummified and scaled the string type fields using a standard scaler method.

Based on the application and research with this dataset, the optimal model is a gradient boosting classifier (GBC). GBC is a random forest where each successive tree collaborates to build the best final model. This model is ideal for datasets with complex interactions between predictors as it tends to return high accuracy from its progressive learning rate method.

I investigated and compared a decision tree (DT), random forest (RF), GBC, an artificial neural network (ANN), and support vector machine (SVC) models. These models yield cross-validated (CV) accuracy scores of .70 for DT, .62 for RF, .74 for GBC, .72 for ANN, and .64 for SVC. The GBC model gave the highest and most reliable accuracy score. CV helps to avoid overfitting from split-out test and training data, and hyperparameter tuning can generate.

1.2. Business Applications

There are two primary business applications of the prediction model. First, for entrepreneurs, the predictive power of the model allows them to predict the success of several projects so that they can focus on ideas that are more likely to succeed – thus leading to successful campaigns, and less time spent on projects that will likely fail. Entrepreneurs can dedicate their limited resources to projects with greater chance of success, instead of those which are more likely to fail. Which could be a game changer for an entrepreneur breaking into the market.

Second, Kickstarter itself can use this model to grow its own revenues. Their business is based on a symbiotic relationship between itself, entrepreneurs, and the community of backers. Kickstarter takes 5% stake in successful projects and is therefore incentivized to grow the number of successful projects to grow their own revenues.[1] Even without the stake, successes help Kickstarter's reputation as the go-to tool for crowdfunding via recency and halo biases. Sharing prediction insights allow Kickstarter to empower entrepreneurs with tips on success and lessons from failed projects. By arming creators with success tips, Kickstarter could grow the number of successful projects, which in turn leads to increased revenues, the ability to empower its community, and ultimately maintain its growth for many years.

2.1. Preprocessing & Model Selection

After removing the unique predicters an iterative clustering and feature selection methodology helped to determine the features to keep. I added a *relative_goal* which is the order of magnitude ($\log_{10}$) of the *goal_inUS*. A K-Nearest neighbor (KNN) model clustered the data, while a shadow score evaluated the accuracy. Initial models split clusters into 2 groups successful and unsuccessful projects. The algorithm removed predictors progressively to uncover deeper levels of insights that maintained high (>0.8) shadow scores. *Category, country, deadline, state, staff_pick backers*

[1] Cuofano, G. How Does Kickstarter Make Money? The Kickstarter Business Model In A Nutshell. *FourWeekMBA* **2022**.

count, blurb_len,Relative_goal and all *launched* related variables were kept based on this method. The final clustering model was a KNN model with 6 clusters with a shadow score of 0.88. 5 of the 6 clusters grouped successful projects together while the last one grouped unsuccessful project.

2.2. Business Applications

Entrepreneurs rely on backers for project funding. Both spotlight and staff picks bring attention to projects, essentially acting advertising, to broadcast the project to the world. The model clusters staff picks or spotlights, and the majority of successful projects together. Almost all failed projects had no spotlight or staff pick. Further, the clusters indicate the timing of campaigns was not a differentiator of staff pick or success except for hardware which was trendy in 2015. Projects that fit into entertainment categories like shorts, or plays were often ignored by staff and backers as they were overrepresented in the failed group. Only hardware and web projects were grouped with success. Lastly, the clusters indicate that projects outside the US typically fail and do not receive a staff pick. Therefore, as an entrepreneur, the keys to success are *staff pick*, *spotlight*, being based in the US, and having a hardware or web-type project.

Kickstarter can learn from cluster insights. The revenue model relies on the number of successful projects. They, therefore, require a strong investor base and project success rate. Staff have a role to play as they can promote projects through their picks, which almost always results in success. However, data suggests that the staff are biased - exclusively picking projects in the US. If staff promote projects outside the US, this will increase project diversity and cause an increased investor diversity via proximity bias. Increasing staff pick diversity would therefore help increase funding diversity (decreasing revenue risk) and avoid US consumer fatigue by promoting new non-US backers and entrepreneurs to participate more. Per the cluster data, Kickstarter has a golden opportunity to grow revenue, their community, empower their employees, expand internationally, and reduce platform bias by adjusting how they pick projects to feature.